

1 Solution — GIAM 1.4

Problem 3

1.1 Problem

The *octal* representation of an integer uses powers of 8 in place notation. The digits of an octal number run from 0 to 7; one never sees 8's or 9's. How would you represent 8 and 9 as octal numbers? What octal number comes immediately after 777_8 ? What (decimal) number is 777_8 ?

1.2 Answers at a Glance

$8 = 10_8,$	$9 = 11_8,$	$777_8 + 1 = 1000_8,$	$777_8 = 511.$
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1.3 Representing 8 and 9 in Octal

Octal has only the digits 0, 1, 2, 3, 4, 5, 6, 7. Counting upward, once you pass 7 there is no single digit left, so — exactly as decimal rolls 9 over to 10 — octal rolls 7 over to 10. The "1" is now a *count of eights*:

$$8 = 1 \cdot 8 + 0 = 10_8, \quad 9 = 1 \cdot 8 + 1 = 11_8.$$

So 8 and 9 — which *cannot* be written with a single octal digit — become the two-digit octal numbers 10_8 and 11_8 .

1.4 What Comes Immediately After 777_8

Every digit of 777_8 is already the maximum octal digit, 7. Adding 1 forces a carry that cascades all the way along — just like adding 1 to 999 in decimal:

$$777_8 + 1 = 1000_8.$$

(The rightmost 7 becomes 0 and carries; the middle 7 becomes 0 and carries; the leading 7 becomes 0 and carries into a new digit 1.)

1.5 777₈ in Decimal

Expand using powers of 8 in place notation:

$$777_8 = 7 \cdot 8^2 + 7 \cdot 8^1 + 7 \cdot 8^0 = 7 \cdot 64 + 7 \cdot 8 + 7 \cdot 1 = 448 + 56 + 7 = 511.$$

So $777_8 = 511$. (And consistently $1000_8 = 1 \cdot 8^3 = 512 = 511 + 1$, confirming the rollover.)

1.6 All Three at Once

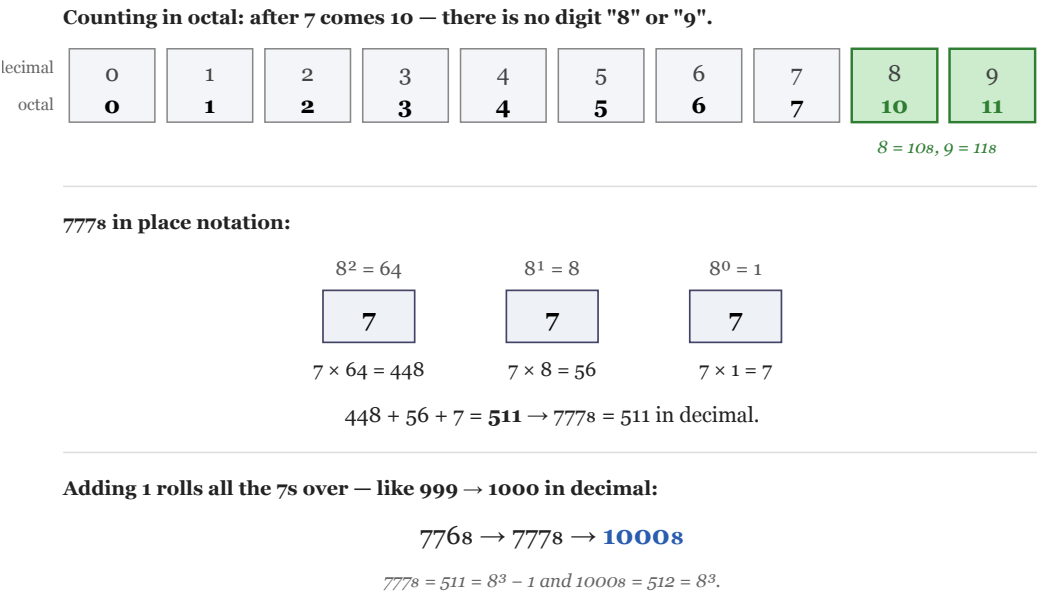


Figure 1.1: Top: counting in octal — decimal 8 and 9 become 10₈ and 11₈ (no single digit for 8 or 9). Middle: $777_8 = 7 \cdot 64 + 7 \cdot 8 + 7 \cdot 1 = 448 + 56 + 7 = 511$. Bottom: $777_8 + 1 = 1000_8$, mirroring $999 \rightarrow 1000$; and $777_8 = 511 = 8^3 - 1$, $1000_8 = 512 = 8^3$.

1.7 Remark — 777₈ = 8³ − 1, the Octal "999"

The rollover $777_8 \rightarrow 1000_8$ is the octal twin of $999 \rightarrow 1000$. In *any* base b , the largest three-digit number has every digit equal to $b - 1$, and it equals one less than b^3 :

$$\underbrace{(b-1)(b-1)(b-1)}_{\text{base } b} = b^3 - 1.$$

- Decimal ($b = 10$): $999 = 10^3 - 1 = 1000 - 1$.
- Octal ($b = 8$): $777_8 = 8^3 - 1 = 512 - 1 = 511$.

This is why 777_8 comes out to the memorable $511 = 512 - 1$, and why the *next* number, $1000_8 = 8^3 = 512$, opens a fresh power-of-8 place.

1.8 Remark — Octal Is Compressed Binary

There is a beautiful link to Problem 2's binary theme: because $8 = 2^3$, **each octal digit corresponds to exactly three binary digits**. The octal digits and their 3-bit forms:

octal	0	1	2	3	4	5	6	7
binary	000	001	010	011	100	101	110	111

Table 1.1

To convert, just translate each octal digit to its 3-bit block. So $7 = 111_2$ gives

$$777_8 = \underbrace{111}_7 \underbrace{111}_7 \underbrace{111}_7_2 = 11111111_2 = 2^9 - 1 = 511,$$

nine 1's in a row — confirming $511 = 2^9 - 1$ from the binary side. Likewise $1000_8 = 001\,000\,000\,000_2 = 2^9 = 512$. Octal is, in effect, *binary written three bits at a time* — which is exactly why it was historically popular in computing, as a compact human-readable stand-in for long bit strings.

1.9 Remark — Why Octal Has No 8 or 9

The number of distinct digits a base needs is exactly the base itself: base b uses the b digits $0, 1, \dots, b - 1$. Octal ($b = 8$) therefore uses 0 through 7 — eight symbols, stopping right before 8. The symbols "8" and "9" are simply *not part of the octal alphabet*; the *quantities* eight and nine still exist, of course, but they must be spelled with multiple

digits (10_8 and 11_8). Conversely, bases *larger* than 10 must *invent new digits*: hexadecimal ($b = 16$) borrows A, B, C, D, E, F for ten through fifteen, since the familiar ten symbols run out.

1.10 Remark — A Quick Sanity Check via Subtraction

A fast way to confirm $777_8 = 511$ without multiplying: 777_8 is “one less than 1000_8 ,” and $1000_8 = 8^3 = 512$, so $777_8 = 512 - 1 = 511$. This $b^3 - 1$ shortcut beats the place-by-place expansion and generalises: $7777_8 = 8^4 - 1 = 4095$, $77_8 = 8^2 - 1 = 63$, and so on. Recognising a string of maximal digits as “(one less than the next power of the base)” is a handy trick in any base.